

ZION TECH HUB DATA CHALLENGE

Where the Budget Should Go: A Data-Driven Ad and Growth Strategy for Zion Tech Hub

A registration-data analysis built for Zion Tech Hub's marketing lead, on two real, unfiltered cohort exports.

1,272

registrations reconciled

46

duplicates resolved

9

assumptions documented

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[Github repository](#)

Three questions, answered up front

Full evidence and detail for each answer follows in the slides ahead.

01

Where should the ad budget go, and on whom?

X (Twitter) 55%, LinkedIn 30%, Facebook/Instagram test 10%, Referral 5%. Targeted by audience: broad Student/Tech/Business reach on X, Healthcare-interest on LinkedIn.

02

What changes grow the community and fill Cohort 11?

Seven specific moves, from clarifying course selection to a deliberate deadline push to a LinkedIn referral program (full list ahead).

03

Where and how should Zion Tech Hub grow outside Nigeria?

Kenya. The only one of the four consistent international markets that grew between cohorts, reachable through the same X and LinkedIn channels.

Two messy exports, turned into one defensible dataset

1,272

registrations reconciled across Cohort 9 and Cohort 10

46

duplicate registrations resolved via a phone + email match

422

spike-driven rows isolated and adjusted for

9

documented assumptions behind every judgment call

4

channel, geography, timing and course dimensions compared

5

deliverables: findings, recommendations, assumptions, and two dashboards

Clean first, then let the two cohorts talk to each other

1

Inspect

Understand what each cohort's export actually contains, field by field, before touching a single value.

2

Clean

Standardize countries and channels, resolve duplicates with a phone-and-email composite key, and flag, not guess, ambiguous entries.

3

Compare

Set Cohort 9 against Cohort 10 on channel, geography, timing and course, controlling for spikes and catalog changes.

4

Recommend

Turn the patterns that survive scrutiny into a specific budget split and a numbered growth plan.

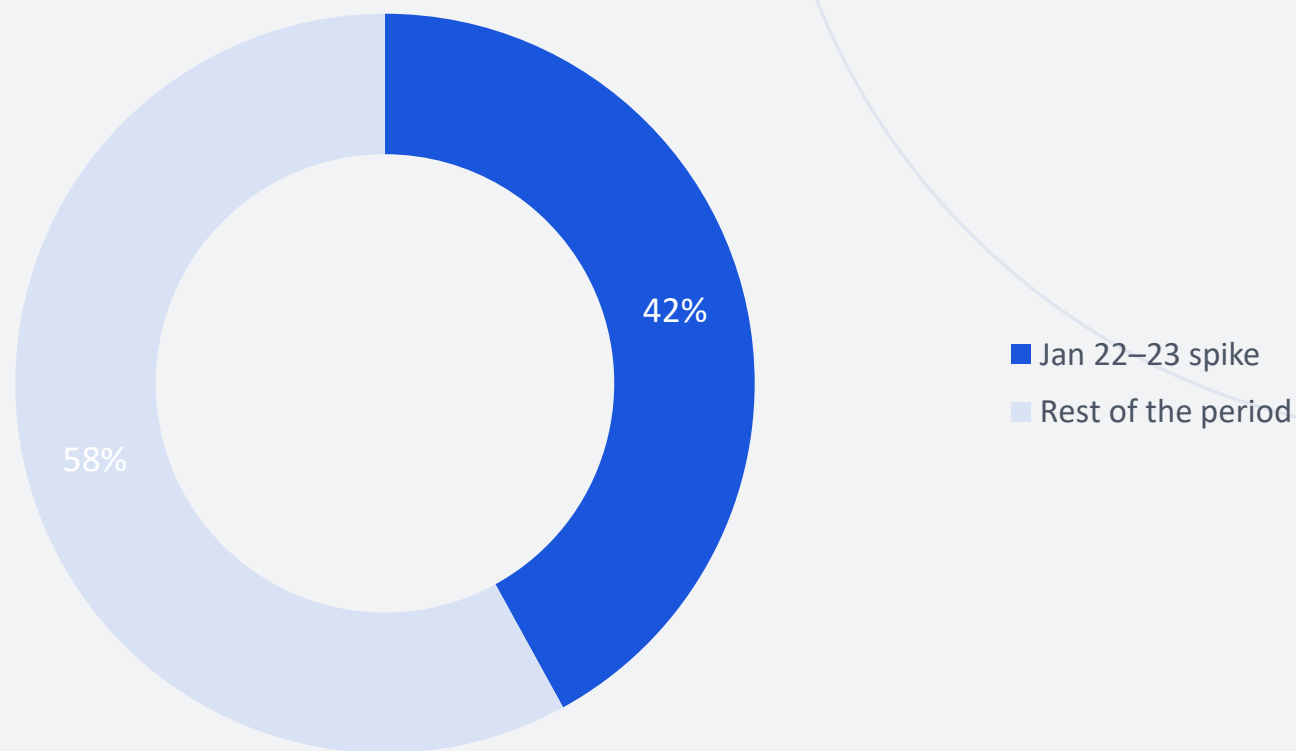
● FINDING ONE

A single two-day spike was inflating the picture

422 Cohort 9 rows arrived through X (Twitter) in a 48-hour window, January 22 to 23. Left in, X looks dominant everywhere; removed, every channel's real performance becomes visible.

42%

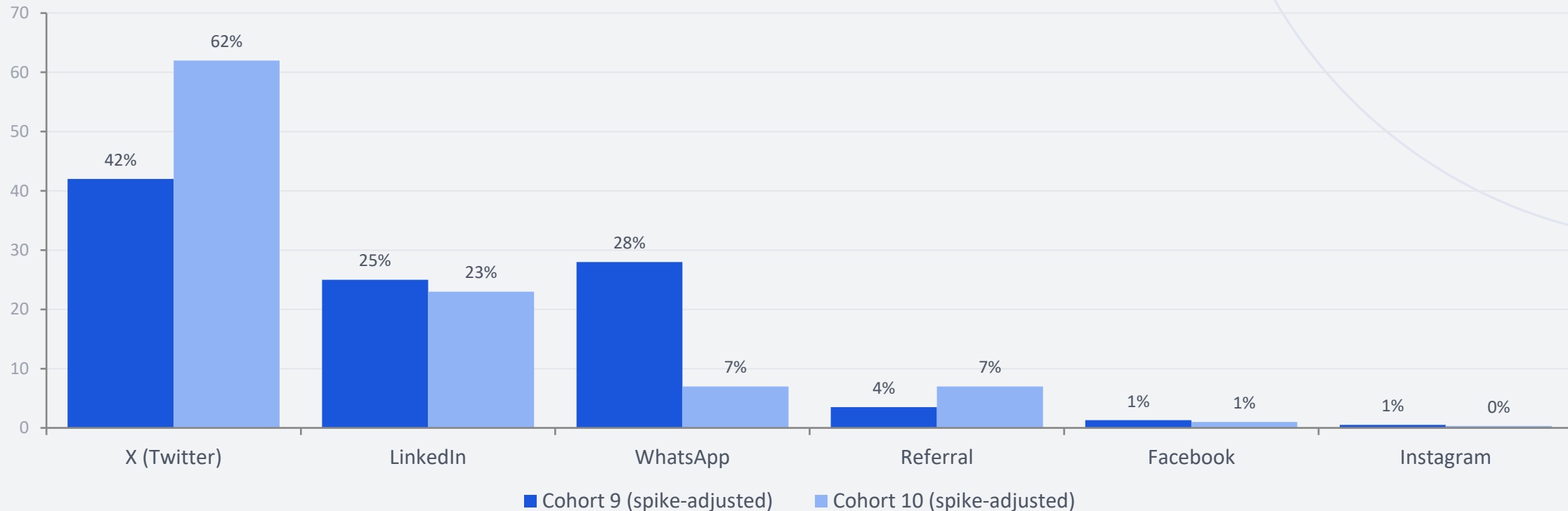
of all Cohort 9 registrations arrived in that single 48-hour window



● FINDING TWO

LinkedIn holds steady. WhatsApp fell hard.

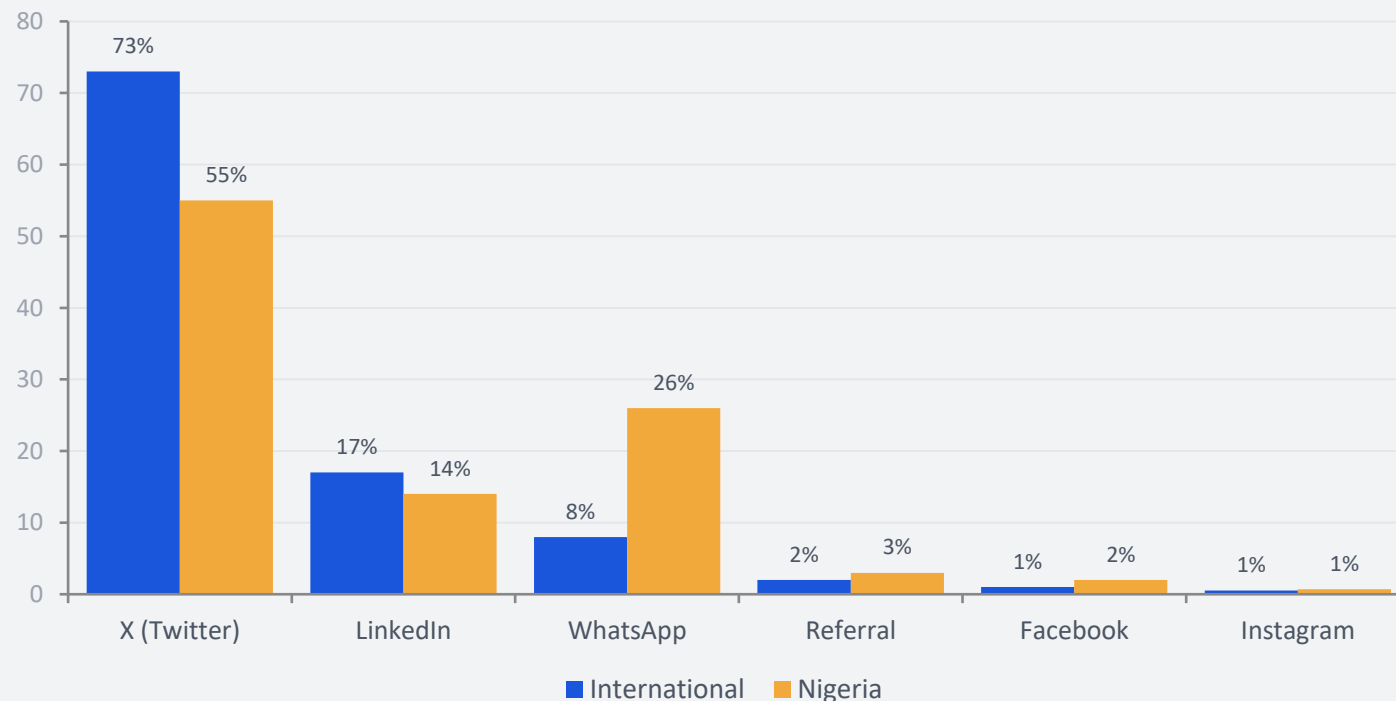
Spike-adjusted share of registrations by channel. X (Twitter) is the only channel with genuine growth between cohorts; WhatsApp drops from 28.9% to 7.0%, too steep to be routine.



● FINDING THREE

Growth outside Nigeria already has a shape

Channel mix by geography, Cohort 9. X (Twitter) and LinkedIn, not WhatsApp, are what actually reach registrants outside Nigeria; Nigeria, South Africa, Ghana and Kenya are the consistent top four across both cohorts.



THE PICK

Kenya

8.2% → 11.9%

share of cohort, Cohort 9 to Cohort 10

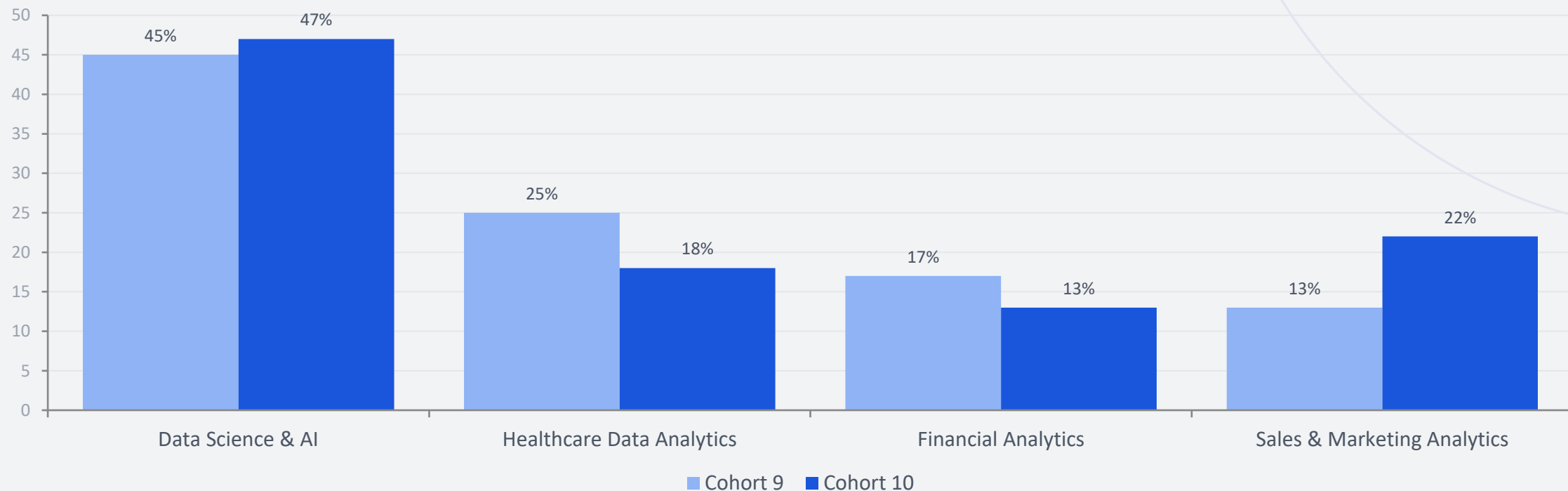
The only one of the four consistent international markets that grew, while South Africa and Ghana both fell, Ghana from a tie for first to nearly last.

Same 55/30 channel split applies, layered with Kenya-specific targeting.

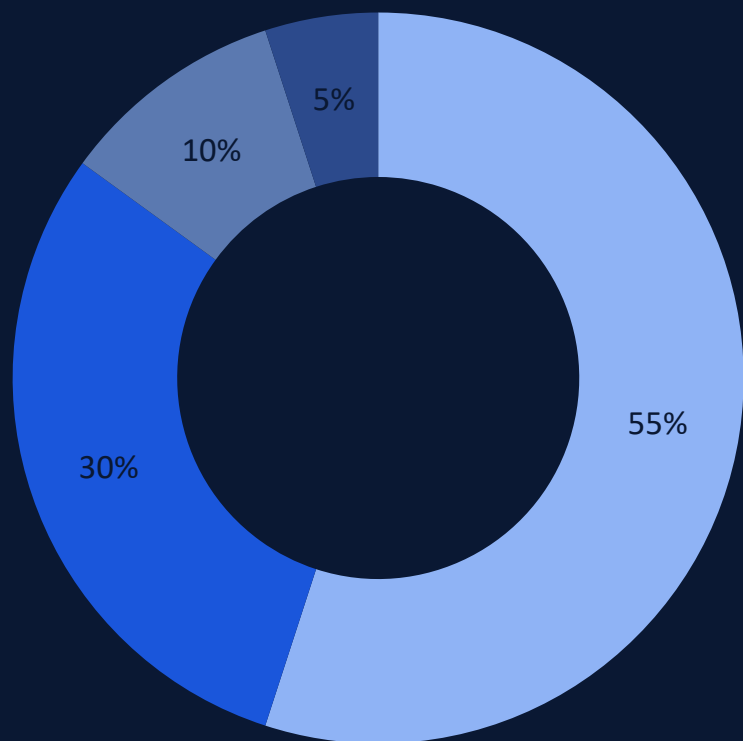
● FINDING FOUR

Healthcare Data Analytics is cooling, even after controlling for dilution

Cohort 10 added new courses, so raw share comparisons are misleading on their own. Controlled for that dilution, Healthcare Data Analytics is still down 6.7 points, and 64.9% of repeat registrants switched their course choice.



How to split the first ad budget



- **X (Twitter)** 55%
Only channel with genuine spike-adjusted growth. Strongest for international reach.
- **LinkedIn** 30%
Most stable channel. Disproportionately strong for Healthcare Data Analytics.
- **Facebook / Instagram (test)** 10%
Vanished in Cohort 10. A small test answers cheaply whether that was real.
- **Referral incentive** 5%
Small, steady organic signal already present, worth actively encouraging.

WHO THIS REACHES

X: Students/Graduates, Tech/Data and Business audiences, 45% of Cohort 10, a broad net, not one profession. LinkedIn: skews Healthcare-interested, 37% vs 19% on X. International spend prioritizes Kenya, the only growing market among the top four. No budget goes to WhatsApp, given its steep spike-adjusted decline.

Seven moves for Cohort 11

1

Clarify course selection upfront

64.9% of repeat registrants switched courses. A short guide would cut this friction.

2

Investigate the WhatsApp decline

28.9% to 7.0% is too steep to ignore. Understand why before doubling down.

3

Plan a deliberate deadline push

Both cohorts convert heavily near the close date. Build this in on purpose.

4

Build a referral program on LinkedIn's base

LinkedIn's stability suggests a loyal audience. Turn it into active referrals.

5

Look into the Healthcare Data Analytics decline

Down 6.7 points even after controlling for new courses.

6

Prioritize Kenya, reached via X and LinkedIn, not WhatsApp

The only one of the top four international markets that grew between cohorts.

7

Lean into the founder's personal brand on LinkedIn. LinkedIn's skew toward Healthcare Data Analytics likely reflects the CEO's own background in the field. More story-sharing there compounds this strength at little to no extra ad spend.

Every judgment call, made visible

Missing names kept, not dropped

55 Cohort 9 rows had a 4-character email pattern with zero variation which is a deliberate redaction, not broken data.

Duplicates required phone AND email

Phone alone falsely flagged 729 rows; short de-identified numbers collide by chance. Requiring both found the real 46.

Ambiguous entries labeled, never guessed

Under 1% of Country values could not be safely resolved. They're marked and excluded rather than forced into a guess.

Course catalog not force-matched

Cohort 10 added two new courses. "Data Science and AI" and "Data Science and ML" were kept separate, not assumed identical.

- EXPLORE IT YOURSELF

A live, interactive dashboard

Filter by cohort, toggle spike days on or off, and watch the channel and geography findings update in real time.

public.tableau.com/app/profile/kofoworola.agbede

Every number in this deck traces back to a decision documented in the project's assumptions writeup so nothing here is a black box.

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Thank you.
Questions and pushback
welcome.